

$$1. \quad \frac{x-5}{4} = \frac{y+2}{-2} = \frac{z+6}{5} \quad \text{pt } (0, 6, 0)$$

$$\text{V.E.: } \vec{r} = (0, 6, 0) + t(4, -2, 5), t \in \mathbb{R}.$$

$$\begin{aligned} \text{P.E.: } x &= 4t \\ y &= 6 - 2t \\ z &= 5t \end{aligned}$$

$$2. \quad xy \text{ plane} \Rightarrow z = 0$$

$$x = 12 - 3t$$

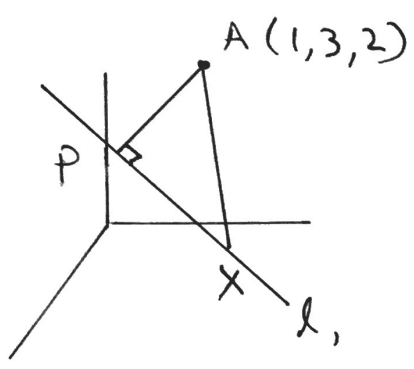
$$y = -8 + 4t$$

$$z = -4 + 2t$$

$$\text{But } z = 0 \Rightarrow -4 + 2t = 0 \therefore t = 2$$

$$\therefore \text{P.O.I is } (6, 0, 0)$$

3.



$$X = (1, 3, 7)$$

$$\vec{AX} = [0, 0, 5]$$

$$\text{Proj}_{\vec{PX}} \vec{AX} = \frac{\vec{AX} \cdot \vec{PX}}{|\vec{PX}|}$$

$$= \frac{[0, 0, 5] \cdot [-1, 1, 2]}{\sqrt{6}}$$

$$= \frac{10}{\sqrt{6}}$$

$$|\vec{AP}|^2 + |\vec{PX}|^2 = |\vec{AX}|^2$$

$$|\vec{AP}|^2 + \frac{100}{6} = 25$$

$$|\vec{AP}|^2 = \frac{25}{3}$$

$$|\vec{AP}| = \frac{5}{\sqrt{3}} \text{ units}$$

(3)

4.

$$V. \in.: \vec{r} = [3, -1, 2] + p[4, 0, 1] + q[2, 0, 1]$$

$$p, q \in \mathbb{R}.$$

p. \in.:

$$x = 3 + 4p + 2q$$

$$y = -1$$

$$z = 2 + p + q.$$

$$5. \pi_1: x + 2y - z + 3 = 0 \quad P(2, -3, 2)$$

$$\vec{OP} = [2, -3, 2]$$

$$\vec{n}_1 = [1, 2, -1]$$

$$\begin{array}{r} 2 \quad -3 \quad 2 \quad 2 \quad -3 \quad 2 \\ \times \quad \times \quad \times \\ 1 \quad 2 \quad -1 \quad 1 \quad 2 \quad -1 \end{array}$$

$$= [-1, 4, 7]$$

$$\vec{OP} \times \vec{n}_1 = [2, -3, 2] \times [1, 2, -1]$$

$$= [-1, 4, 7]$$

$$\therefore -x + 4y + 7z = 0 \quad (\text{OR}) \quad x - 4y - 7z = 0$$

$$\uparrow$$

$$D = 0 \text{ since}$$

$$\text{origin is on}$$

$$\text{the plane}$$

6. $2x - 6y + 4z + 3 = 0$

$$3x - 9y + 6z + k = 0$$

If $k = \frac{9}{2}$, then the 2 planes are

parallel and coincident.

If $k \neq \frac{9}{2}$, then the 2 planes are

parallel and distinct.

Hence, they will never intersect.

7. $P(3, 0, -4)$

$$\pi_1: x + 2y - 7z - 3 = 0 \rightarrow \vec{n}_1 = [1, 2, -7]$$

$$\pi_2: x - 5y + 4z - 1 = 0 \rightarrow \vec{n}_2 = [1, -5, 4]$$

$$\vec{n}_1 \times \vec{n}_2 = [1, 2, -7] \times [1, -5, 4]$$

$$= [-27, -11, -7]$$

$$\begin{array}{ccc} 1 & 2 & -7 \\ \times & \times & \times \\ 1 & -5 & 4 \end{array}$$

$$= [-27, -11, -7]$$

$$\therefore -27x - 11y - 7z + 0 = 0$$

$$-27(3) - 7(-4) + 0 = 0$$

$$D = 53$$

$$\therefore 27x + 11y + 7z - 53 = 0$$

$$8. \quad x + 3y + 3z = 8 \quad (1)$$

$$x - y + 3z = 4 \quad (2)$$

$$2x + 6y + 6z = 16 \quad (3)$$

$$\left. \begin{array}{l} 2\vec{n}_2 = \vec{n}_1 \\ 2D_2 = D_1 \end{array} \right\} \Rightarrow (1) \text{ \& } (3) \text{ are parallel and coincident.}$$

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(2) is not parallel to (1) or (3)

Hence, (1), (2) & (3) intersect in a line.

Use (1) and (2)

Eliminate x: (1) - (2)

$$\begin{array}{r} x + 3y + 3z = 8 \\ - (x - y + 3z = 4) \\ \hline 4y = 4 \end{array}$$

$$4y = 4$$

$$\boxed{y = 1}$$

$$\text{let } z = t$$

$$\Rightarrow x = 5 - 3t$$

$$y = 1$$

$$z = t$$

Eliminate y: (1) + 3(2)

$$\begin{array}{r} x + 3y + 3z = 8 \\ + 3(x - y + 3z = 4) \\ \hline 4x + 12z = 20 \end{array}$$

$$4x + 12z = 20$$

$$x + 3z = 5$$

$$x = 5 - 3z$$

(6)

$$9. \quad \vec{r} = [4, 0, 3] + s[-8, 1, -9] + t[-1, 5, 7] \quad (1)$$

$$\vec{p} = [-14, 12, -1] + u[1, 1, 3] + v[-2, 1, -1] \quad (2)$$

$$\vec{n}_1 = [-8, 1, -9] \times [-1, 5, 7] \quad \begin{array}{ccc} -8 & 1 & -9 \\ -1 & 5 & 7 \end{array} \begin{array}{c} \times \\ \times \\ \times \end{array} \begin{array}{ccc} -8 & 1 & -9 \\ -1 & 5 & 7 \end{array}$$

$$\vec{n}_1 = [52, 65, -39]$$

$$\vec{n}_2 = [1, 1, 3] \times [-2, 1, -1] \quad \begin{array}{ccc} 1 & 1 & 3 \\ -2 & 1 & -1 \end{array} \begin{array}{c} \times \\ \times \\ \times \end{array} \begin{array}{ccc} 1 & 1 & 3 \\ -2 & 1 & -1 \end{array}$$

$$\vec{n}_2 = [-4, -5, 3]$$

$$\therefore \vec{n}_1 = -13 \vec{n}_2 \Rightarrow \text{planes are parallel,}$$

distinct or coincident?

$$-4x - 5y + 3z + D = 0 \quad \text{sub. } (-14, 12, -1)$$

$$-4(-14) - 5(12) + 3(-1) + D = 0$$

$$56 - 60 - 3 + D = 0$$

$$D = 7$$

$$\therefore -4x - 5y + 3z + 7 = 0 \quad (3)$$

sub. pt. from (1) into (3) : use (4, 0, 3)

$$-4(4) - 5(0) + 3(3) + 7 = 0$$

$$-16 + 9 + 7 = 0$$

$$0 = 0 \Rightarrow \text{coincident}$$

$\therefore$ planes are parallel and coincident.